

系統程式設計

Lab7 setjmp/longjmp

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1. Where are the errors located?

compare_at() checks for @ and longjmp 1 if no @s are found or longjmp 2 if more than 1 @ is found.

compare_sharp() checks for # in the string and longjmp 3 if any # is found.

compare_asterik() checks for * in the string and longjmp 4 if any * is found.

compare_doubleDot() checks for .. in the string and longjmp 5 if any .. is found.

compare_dot() checks for any . after @ and longjmp 6 if no . is found after @.

The error's location will be shown in the msg, counting 1 from the leftmost character.

2. What is the error message format?

(you must not print any message after you print error message)

I've made a void print_error function that accepts a char ptr and prints "Email format error: " then the string. Using switch-case I can differentiate different errors.

1: Email format error: at least 1 @

2: Email format error: more than 1 @

3: Email format error: # is not allowed

4: Email format error: * is not allowed

5: Email format error: .. is not allowed

6: Email format error: at least 1 . after @

3. How did you use the setjmp/longjmp function?

In my program I use them like try catch error handling. In main setjmp is called and it returns 0 bypassing the switch-case block and proceeds to read file. The program calls the checking functions in a nested fashion, when an invalid format is spotted at any level, longjmp is called and sets jmpbuffer to corresponding error number, that way the switch block can detect which error occurred and print out the correct error msg.

4. Final result.

Please attach screenshot !

```
lk@LK Lab7 % cd "/Users/lk/Documents/GitHub/CCU-CS-ab7_stu"
-----
spta@embedded.cs.ccu.edu.tw
@ found at 5
valid email address!
-----
twetrewt.terthbbb.com.tw
Email format error: at least 1 @
-----
spta@embed@ded.cs.cc@u.edu.tw
@ found at 5
@ found at 11
@ found at 21
Email format error: more than 1 @
-----
pv#pig@hotmail.com
@ found at 7
# found at 3
Email format error: # is not allowed
-----
pvpig@hotmail*.com
@ found at 6
* found at 10
Email format error: * is not allowed
-----
aaawww@ccu.cs..edu.tw
@ found at 7
.. found at 14
Email format error: .. is not allowed
-----
EA.EA@ccucsedutw
@ found at 6
Email format error: at least 1 . after @
```